

(Robust) Deep Hedging

Pietro Amin Puddu

Abstract. Building on seminal continuous-time frameworks, traditional finance calibrates parametric models (local/stochastic volatility) to market prices to derive hedges. In complete markets, where risk factors match hedging instruments, perfect replication is possible; in incomplete markets, practitioners must instead utilize risk-minimizing strategies.

1. Introduction to Hedging

1.1. Greek Hedging

The standard hedging methodology in derivative finance is based on the assumption of a complete market. Computing the matrix ΔH_0 (i.e. the day-zero first order sensitivities of the hedging instruments, e.g. Delta, Vega, Rho, etc.) and the row vector ΔZ_0 (i.e. the day-zero first order sensitivities of the hedged instrument(s)), and then finding the hedging policy δ which "flattens the greeks"

$$\|\Delta Z_0 + \delta \cdot \Delta H_0\|_2$$

Unfortunately, there is no obvious methodologically sound approach to balance between cost and residual risk since a given greek exposure has no dimensionality which can be compared to the cost of entering into it. Formally, this is addressed by using "Cash Greeks" $\Delta_{\$} := x \cdot \Delta$

$$\inf_{\delta} \left\{ \hat{H}(\delta) + \|\Delta^{\$} Z_0 + \delta \cdot \Delta^{\$} H_0\|^2 \right\}$$

1.2. Parameter and Statistical Hedging

The next advancement in hedging strategy involves recognizing that the return vector, dx_0 , is inherently uncertain. This uncertainty—captured in the statistical covariance matrix Σ —should dictate which Greeks are hedged and to what extent. Specifically, if two risk factors present identical Delta exposures and hedging costs, the factor with higher volatility should be hedged more aggressively. Using a second-order Taylor expansion, we approximate the return of any instrument F (where $F \in \{Z, H_1, \dots, H_n\}$) as $dF_0 = \Delta F_0 dx_0 + \frac{1}{2} \Gamma F_0 \Sigma \frac{dx_0^2}{dt} dt + F_0 \Theta dt$. Given the above, it is very tempting to approximate dx_0 itself as a normal with mean $E[dx_0]$ and variance Σ , which means both dZ and dH are approximated as normals — integrate this into a Markowitz mean-variance framework:

$$\min_{\delta} \left[\text{Cost}(\delta) - \underbrace{\mathbb{E}[dZ + \delta dH]}_{\text{Gamma-Theta}} + \lambda \text{Var}(dZ + \delta dH) \right]$$

However, the implicit assumption that all derivatives can be modelled over $[0; \tau]$ as *normals* breaks down in situations with high gamma values, for examples when working with barriers products or short-term options. For these complex scenarios, one must move beyond Taylor approximations and use the actual historical or simulated returns of the instruments, i.e. *Statistical hedging*.

2. Introduction to Deep Hedging

Deep hedging employs policy gradients to optimize neural networks (NNs) toward a specific hedging objective. While NNs can approximate optimal strategies in complex environments—incorporating frictions like transaction costs or vega limits that defy analytic solutions—the approach is not truly model-free. The agent is intrinsically tied to the "market generator" used for training, making its efficacy dependent on that presumed environment.

$$p(Z) = \inf_{\delta \in \mathcal{H}} \rho \left(-Z + \sum_{t=0}^{T-1} \delta_t (S_{t+1} - S_t) - C_T \right) - \pi(0) = \pi(-Z) - \pi(0)$$

Here, $\pi(-Z)$ represents the minimum initial capital required to render the liability $-Z$ acceptable under the chosen risk measure (the replication cost). The indifference price $p(Z)$ scales this cost against $\pi(0)$, the baseline risk of the hedger's "zero-liability" state. This adjustment is crucial: if a risk-neutral or statistical arbitrage opportunity exists, $\pi(0)$ may be non-zero, requiring the price to account for the hedger's alternative risk-adjusted gains.

3. Relationship between deep hedging and delta hedging

Why does the Knightian Uncertainty of the mean matter? Finding an optimal hedging strategy is very sensitive to the mean. Machine thinks it can make money ... starts to shorting vanilla puts and long delta — and the issue lies in the fact that the mean drift is the estimation error for using the input data.

$$U(X) = E_Q[X] + \underbrace{U(X - E[X])}_{\text{Risk term}}$$

Deep hedging can be skewed by statistical arbitrage opportunities arising from a positive drift in the price process of hedging instruments. Specifically, the model may prioritize profits from trading these assets instead of focusing on the payoffs from contingent claims, combining delta hedging and statistical arbitrage — i.e. deep hedging can be hampered by the existence of statistical arbitrage. This decomposition, $\delta^{DH} = \delta^{SA} + \Delta$, implies that deep hedging is inherently hampered by market inefficiencies under the real-world measure $\mathbb{P}$. A statistical arbitrage strategy satisfies $\rho((\delta^{SA} \cdot S)_T) = \pi(0) < 0$. Conversely, if no such opportunity exists ($\pi(0) \geq 0$), the deep hedge coincides with the delta hedge, and the indifference price simplifies to $p(Z) = \pi(-Z)$.

$$\sup_{\delta} \mathbb{E}^* \left[\sum_t \delta_t \cdot \Delta H_t - c_t(\delta_t) \right] \leq 0$$

Under $\mathbb{Q}^*$ the market has no statistical arbitrage opportunities in the sense that there is no strategy a which has positive expected returns. In the absence of transaction costs, $\mathbb{Q}^*$ is an equivalent martingale measure. In the presence of transaction costs, we show that removing statistical arbitrage is equivalent to the measure being an equivalent near-martingale measure, in the sense that the drift of all tradable instruments must be dominated by the transaction costs. Taking the limit to small transaction cost.

3.1. Hedging spread sensitivities

Analyzing the relative spread $\frac{\|\delta^{DH} - \Delta\|}{\|\Delta\|}$ reveals key sensitivities: The gap between deep and delta hedging increases linearly with the asset's drift, while it decreases non-linearly as volatility increases (where market noise drowns out the drift signal). The statistical arbitrage component exhibits a non-linear, inverse relationship with absolute stock price levels. While *robust risk measures* (e.g. $\text{CVaR}_{0.5}$) can mitigate sensitivity to drift estimation, deep hedging under $\mathbb{P}$ requires precise drift calibration. Because derived strategies only align when robust to statistical arbitrage, interpreting resulting prices remains challenging. Consequently, it is often more practical to employ risk-neutral valuation for the initial price, subsequently optimizing the hedge based on the resulting P&L distribution including transaction costs.

3.2. Transitioning to Near-Martingale Dynamics [2]

The solution works as a Girsanov transformation, such that given the optimal trading strategy the change of measure removes such statistical arbitrage — leveraging into the derivative of the utility function. The utility-based construction of the risk-neutral density D^* is model-agnostic. It relies solely on solving the optimization of a^* and y^* without specifying underlying dynamics for $\mathbb{P}$. By numerically solving the stochastic control problem on N simulated paths, we can reweight the sample set so that the resulting weighted distribution is a (near-)martingale. D^* acts as a customized path-filter. By setting the weight of each path to the marginal utility $u'(\dots)$, the latter decreases as wealth increases, paths where the optimized strategy (a^*) made an easy profit from drift get a lower weight.

4. Introduction to Robust Deep Hedging

Deep Hedging is not automatically robust to misspecification of the training model or changes in the market environment (distributional shift). This can be very concerning, significantly when the option to-be-hedged contains a forward-starting component.

4.1. Ambiguity-Adverse Deep Hedging

This approach investigates the use of *smooth ambiguity aversion* particularly in the context of deep hedging, through feature clustering to group paths from a market generator into a predefined number of clusters. These clusters are chosen according to the values of certain path features that are believed by the modeler to affect the choice of effective hedging strategies. These clusters are interpreted as a manifestation of model uncertainty, which allows us to apply ideas from the theory of smooth ambiguity aversion by measuring risk within each cluster by an inner risk measure, and model uncertainty by an outer risk measure over the probability measure defined by the clusters. An *ambiguity-averse* deep hedging strategy is one that minimises this nested risk measure — with no additional training data, and in particular with no modification to the market generator which would otherwise be used for deep hedging, as well as without charging a higher premium than the benchmark. While the canonical choice for ξ is the empirical distribution of clusters, it can be modified for scenario re-weighting.

4.1.1 The Inner Loss Function: Risk Measure

The inner loss function is designed to incorporate transaction costs while penalizing arbitrage-seeking behavior (λ is the risk aversion parameter, which effectively assigns a monetary (dollar) value to units of variance),

$$\rho = \text{Var}(\text{P\&L before transaction costs}) + \lambda \mathbb{E}[\text{Total transaction costs}]$$

4.1.2 The Outer Loss Function: Risk Measure

4.2. Other Attempts at Robustification of Deep Hedging

In [6], the authors propose a market generator for the purposes of robustification of the deep hedging strategy. This generator is a discretely-sampled SDE of the form

$$dX_t = (b_0 + b_1 X_t)dt + (a_0 + a_1 X_t)dW_t$$

in which—crucially—the parameters b_0, b_1, a_0, a_1 and σ are sampled from a uniform distribution at every timestep, independently of every other path and time. The strategies learned by their deep hedgers also may learn a significantly higher price than their benchmark (in the case of the lookback option).

In [7], there are three components to their training procedure: the generator which transforms Brownian increments into the underlying asset price process, the discriminator which is the deep hedging

component, and the penaliser which penalises the market model produced by the generator. However, critics of deep hedging already point to its black-box characteristics, and the added complexity of having a market generator be trained alongside the deep hedging strategy contributes further to this. The generator’s goal is in some sense to provide worst-case scenarios to the deep hedger, though these may not actually be realistic paths. Existing market generators may possess both realistic and non-realistic aspects, and robustifying against both may be too conservative in comparison with only robustifying against the non-realistic ones.

In [8], the maxmin expected utility framework is shown to be the special limiting case of infinite ambiguity aversion in the framework they dub smooth ambiguity aversion. In other words, the maxmin expected utility framework is (in a precise sense) that of an infinitely ambiguity-averse decision maker. This model, *smooth ambiguity aversion*, proves that a preference order on decision making under uncertainty can be characterised by—essentially—having a utility function for risk, and another (“outer”) utility function for uncertainty.

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